

PIA 401 Course Outline

PIA 401 – Public Investment Management Fundamentals

The cycle, the eleven gates, and what a gate is for.

School	04 – Public Investment
Hours	30 (indicative band 27-33)
Version	1.0 (2026-09-05)
Model	AICTEM PIAS, D:/EDIP/models/PIASModelling
Reference country	Utopia (UTP), UTD 2024 prices
Prerequisites	None. PIA 401 is the entry point to School 04 and assumes only that a learner can read a table of money over time. Arithmetic beyond multiplication and division is introduced where it is needed. PIA 403 and PIA 404 assume this course.
Audience	Development committee secretariats, planning units, ministry project officers, budget desk officers, and anyone who has to say why one project was funded and another was not
Pass mark	60 %

What this course is

Public investment management is the set of arrangements by which a government decides what to build, and it fails in a characteristic way. It does not usually fail by choosing a project that a spreadsheet said was bad. It fails by choosing a project that no spreadsheet ever examined, at a stage when the decision had already been announced, on a cost estimate nobody could trace, with a recurrent burden that was never added up.

This course teaches the cycle that is supposed to prevent that, and it teaches it against a system that implements it. AICTEM PIAS holds a project register, eleven appraisal gates, a readiness index, a multi-criteria score and a governed parameter set, and it refuses to produce numbers it cannot justify. You will learn the cycle by walking eleven real appraised projects through it and reading what the system says about each.

The course is deliberately not a cost-benefit course. It does not teach you to compute an ENPV – PIA 404 does that. It teaches you what an ENPV is for, where in the cycle it arrives, what it is allowed to decide, and what the other ten gates are protecting that the economic gate does not.

Its central discipline is that a decision must be reconstructable. A rank on its own is not auditable, so PIAS does not store one: it stores the cohort, the weight set, the weight set version, and every raw and normalised criterion value beside it. This course asks you to reconstruct a rank from those parts, by hand, and it does not let you quote one you cannot.

What a graduate can do

Describe the eight stages of the public investment cycle and say which decision each is protecting; name the eleven appraisal gates, say which eight are blocking and who may waive one; read a project record and identify what has not been done to it; compute a Project Readiness Index by hand, including the renormalisation that a NOT_APPLICABLE element forces; reconstruct a multi-criteria rank from its stored components and state how much of the weight was never scored; explain why a project's rank changes when a different project is added to the cohort; identify the recurrent burden a capital decision commits the budget to in perpetuity; and trace any number in an appraisal report back to the governed parameter, version and effective date that set it.

What this course does not claim

This course does not teach cost-benefit analysis. It shows you an ENPV and tells you what it means; PIA 404 tells you how to compute one and PIA 406 tells you what a conversion factor is. It does not teach financial appraisal (PIA 403), risk (PIA 407 and PIA 408), climate (PIA 409), spatial analysis (PIA 410) or portfolio selection (PIA 411).

It does not supply a national appraisal standard. The discount rates it quotes are recorded in the model's own configuration as "placeholder pending national confirmation", and the course repeats that label every time it uses them. A learner who leaves this course quoting 10.0% as Utopia's social discount rate, without the word placeholder, has misunderstood the most important thing in it.

It does not claim the eleven sample projects are real. They are the model's seeded register. They are realistic, they span eight sectors, three of them have a negative ENPV and one fails its needs assessment – all of which is deliberate. None of them is a project the Government of Utopia is building.

It does not claim that a system prevents bad decisions. PIAS can record that the STRATEGIC gate is PENDING on all eleven projects, which it does. It cannot make anybody assess it. The course is explicit that a gate is a place to put evidence, not a substitute for having any.

Learning outcomes

1. Describe the eight stages of the public investment cycle and say which decision each stage is protecting.
2. Read a project record and state what has and has not been done to it.
3. Explain what a project is when it is data: the register, the versions and the audit trail.
4. Name the eleven appraisal gates, identify the eight that are blocking, and say who may record PASS_WITH_CONDITIONS on one.
5. Read a gate result and distinguish PENDING from FAIL, and both from an absence of evidence.
6. Relate an evidence stage to the optimism bias uplift it attracts, and explain why the uplift falls as the evidence improves.
7. Separate capital, recurrent and lifecycle cost, and identify what a salvage credit rests on.
8. Compute a Project Readiness Index by hand, including the renormalisation a NOT_APPLICABLE element forces.
9. Reconstruct a multi-criteria rank from its stored components and state the unscored weight share.

10. Explain why adding a project to a cohort changes the scores of the others, and why the cohort is therefore stored with the rank.
11. Identify the steady-state recurrent burden a capital decision commits the budget to, and say why it is a headline number.
12. Trace any number in an appraisal report back to the governed parameter, the version and the effective date that set it.

Module structure

Module	Title	Sessions	Hours	Laboratory	Manual	Assessment
M01	What public investment management is	1	1.25	–	1-2	Quiz M01
M02	The cycle, and what each stage protects	1	1.25	–	3	Quiz M02
M03	The project record	1	3.25	Lab 01	4	Quiz M03
M04	The eleven gates	2	3.50	Lab 02	5-6	Quiz M04
M05	Evidence stages and optimism bias	1	1.50	–	7	Quiz M05
M06	Cost: capital, recurrent, lifecycle	1	1.50	–	8	Quiz M06
M07	The appraisal conventions	1	1.50	–	9	Quiz M07
M08	Readiness and the PRI	1	3.50	Lab 03	10	Quiz M08
M09	Scoring, normalisation and rank	2	3.50	Lab 04	11-12	Quiz M09
M10	The recurrent burden	1	1.50	–	13	Quiz M10
M11	Governance: parameters, versions, audit	1	3.50	Lab 05	14	Quiz M11
M12	From appraisal to budget	1	1.50	–	15	Quiz M12

Laboratories

Lab	Title	Hours	Deliverable
1	Read a project record, and list what has not been done	2.0	A one-page absence statement
2	Walk a project through all eleven gates	2.0	An eleven-row gate profile with a licence statement
3	Compute a Project Readiness Index by hand	2.0	Two worked PRI computations
4	Reconstruct a rank, then change the weight set	2.0	A rank reconstruction and a reweighted ranking
5	Trace a number to the parameter that set it	2.0	Five completed traces

Assessment

Component	Weight
Knowledge quizzes	15 %
Laboratory exercises	30 %
Applied assignments	25 %
Capstone / final project	30 %

Applied assignments

ID	Title	Weight	Deliverable
A1	The absence statement	20	Marked on completeness of the absence list and correctness of the parameter resolution
A2	The gate profile and its licence	25	Marked on refusal to infer, and on the honesty of the licence statement
A3	Readiness and rank, reconstructed	30	Marked on arithmetic, on the renormalisation, and on the unscored share
A4	The traced number	25	Marked on the sixth step: standard or placeholder

Final project

A complete decision note on one project from the sample register: its record, its position in the cycle, its status at all eleven gates with the blocking ones identified, its readiness index recomputed by hand from the element statuses, its rank reconstructed from the stored criterion contributions with the unscored weight share stated, its steady-state recurrent burden, and a recommendation that says what would have to be true for the recommendation to change.

Folder contents

Every course in the Academy carries the same section 5 folder. See `README.md` for what each item holds and `COURSE_BUILD_REPORT.md` for what was measured.